

Technology Stack

Date	3 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML5, CSS3, JavaScript</i>	<i>Used to develop a simple, responsive, and interactive web interface for entering soil and environmental parameters and displaying crop recommendation results.</i>

2	Backend / Server-Side	<i>Python 3.10+, Flask</i>	<i>Flask provides a lightweight backend framework for handling user requests, processing input data, loading the trained machine learning model, and generating crop recommendations.</i>
3	Database / Data Storage	<i>CSV Dataset (Crop Recommendation Dataset), Pickle (.pkl) Model</i>	<i>The agricultural dataset is stored in CSV format for model training, while the trained machine learning model is saved as a Pickle (.pkl) file for efficient prediction.</i>
4	Cloud / Hosting / Deployment	<i>Flask Development Server (Local Deployment)</i>	<i>Used for developing, testing, and running the application locally, enabling real-time crop recommendations through a web browser.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Git is used for version control, and GitHub is used to store the project repository, collaborate with team members, and track code changes.</i>

6	Third-Party APIs / Other Tools	<i>Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook</i>	<i>Scikit-learn is used for building and evaluating machine learning models; Pandas and NumPy for data preprocessing; Matplotlib and Seaborn for data visualization; Jupyter Notebook for model development and experimentation.</i>
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